Statistical-zero-knowledge containment from low-error randomized many-one reductions to random strings

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Editorial note. Faithful editable TeX transcription of formal author Theorem39 and its complete AppendixA proof pp29:29-29:30, exact universal-machine/random-string promise definitions and standard symmetry-of-information statement, complete published bibliography. Original author mathematical expressions retained; no corrections or additional proofs. This formal low-error many-one SZK containment is independent from the general nonadaptive AM containment; informal introduction statements are not used as targets.

Allowed standard prerequisites (editorial).

- Unbounded Kolmogorov symmetry of information. Exact author Proposition11 retained.
- Entropy Difference in SZK and closure under polynomial many-one reductions. The complete standard containment fact as stated by the authors in AppendixA is retained in the proof.

Author descriptions and Kolmogorov complexity (Section 2.4). Throughout we assume a fixed pairing function $\langle\cdot\rangle$ and a standard encoding of Turing machines M by binary strings, with $\widetilde{M}$ denoting the encoding of M. We fix a universal Turing machine U that for any Turing machine M and input x takes as input $\langle\widetilde{M}, x\rangle$ and outputs $M(x)$. The Kolmogorov complexity $K(q)$ of q (with respect to U), denoted $K(q)$, is the minimum length of y such that $U(y) = q$. We define the Kolmogorov complexity of a machine M to be $K(\widetilde{M})$.

Proposition 11 (Symmetry of information). Let x, y be strings. Then $K(x; y) = K(x) + K(y \mid x) + O(\log(K(x, y)))$.

Author random-string promise problems (Section 2.5). In the R_K problem, the instance is a string x, and the question is whether $K(x) \geq |x|/2$. Given a function $g : \mathbb{N} \to \mathbb{N}$, the R_K problem with gap g is the promise problem whose NO instances are strings x with $K(x) \geq |x|/2$ and whose YES instances are strings x with $K(x) < |x|/2 - g(|x|)$. Given a function $\alpha : \mathbb{N} \to \mathbb{R}$ such that $\alpha(n) \geq 1$ for all n, the R_K problem with multiplicative gap α is the promise problem whose NO instances are strings x with $K(x) \geq |x|/2$ and whose YES instances are strings x such that $K(x) < |x|/(2\alpha(|x|))$.

Theorem 39. Let L be any decidable language such that there is a randomized poly-time m-reduction with error $1/n^{\omega(1)}$ from L to R_K with gap g for some time-constructible $g = \omega(\log(n))$. Then $L \in \mathsf{SZK}$.

Appendix A. Limitations of Randomized m-Reductions to Approximating Kolmogorov Complexity. In this section, we show that any language that has a low-error randomized polynomial-time m-reduction to a gap version of the Kolmogorov random strings is solvable in Statistical Zero Knowledge.

Proof. Let L be a language as in the statement of the theorem. Let f be a randomized poly-time m-reduction with error $1/n^{\omega(1)}$ to R_K with gap g for some $g = \omega(\log(n))$. We can assume without loss of generality that any query made by f on input x of length n is of length $p(n)$ for some polynomial p. Indeed, let $p(n)$ be an upper bound on the running time of f. We modify the reduction by "padding" any query made by f to length $m = 2p(n)$ as follows. Let c be the constant in the $O(\cdot)$ in Proposition 11. If the query q is of length k, we sample a string $y = r0^{(m-k)/2-2c\log(n)}$ where $r \in \{0,1\}^{(m-k)/2+2c\log(n)}$ is chosen uniformly at random, and output the query qy. Observe that by Proposition 11, with probability $1 - 1/n^{\omega(1)}$, if q is a YES instance of R_K, then qy is a YES instance of R_K, and if q is a NO instance of R_K, then qy is a NO instance of R_K. For each $x \in \{0,1\}^*$, $f(x)$ is a random variable supported on $\{0,1\}^{2p(|x|)}$. We claim that the following holds. If $x \notin L$, the random variable $f(x)$ has entropy at least $p(|x|) - g(2p(|x|))/2 + 1$, and if $x \in L$, the random variable $f(x)$ has entropy at most $p(|x|) - g(2p(|x|))/2 - 1$. We first show that the claim implies that $L \in \mathsf{SZK}$, and then establish the claim. To see that the claim implies $L \in \mathsf{SZK}$, we use the fact that Entropy Difference is in SZK. Namely, there is a statistical zero-knowledge protocol that, given a circuit C sampling a distribution over m bits and a parameter s, accepts when the entropy of C is at least $s+1$ and rejects when the entropy of C is at most $s-1$. Note that a circuit C sampling $f(x)$ can easily be computed from x. Since SZK is closed under poly-time m-reductions, we have that $L \in \mathsf{SZK}$, by reducing to Entropy Difference with parameter $m/2 - g(m)/2$. Next we establish the claim. Assume for the sake of contradiction that the claim fails for infinitely many inputs. Let $\{n_i\}$ be an infinite sequence of numbers such that the claim fails for at least one input of length n_i for each i, and let x_i be the first input of length n_i for which the claim fails. Note that $K(x_i) = O(\log(n_i))$. The reason is that we can describe x_i by a program which has n_i in binary encoded into it, and then searches for the first string z of length n_i such that the claim fails for z, i.e., either $z \in L$ and the entropy of $f(z)$ is greater than $p(|z|) - g(2p(|z|)/2 - 1$, or $z \notin L$ and the entropy of $f(z)$ is smaller than $p(|z|) - g(2p(|z|))/2 + 1$. Let i be large enough

and x_i be the first string of length n_i for which the claim fails. Let $m = 2p(n_i)$. Either $x_i \in L$ and the entropy of $f(x_i)$ is greater than $m/2 - g(m)/2 - 1$, or $x_i \notin L$ and the entropy of $f(x_i)$ is smaller than $m/2 - g(m)/2 + 1$. We show that either case leads to a contradiction. In the first case, the random variable $f(x_i)$ has entropy greater than $m/2 - g(m)/2 - 1$. Then with probability at least $1/m$ over $y \in \{0,1\}^m$ sampled from $f(x_i)$, $K(y) > m/2 - g(m)$. Indeed, if not, the entropy of $f(x_i)$ is at most

$$\frac{1}{m} * m + (1 - 1/m) * (m/2 - g(m)) \leq m/2 - g(m) + 1 < m/2 - g(m)/2 - 1,$$

which contradicts the lower bound on the entropy of $f(x_i)$. But if $K(y) > m/2 - g(m)$ with probability at least $1/m$, the reduction cannot be correct for $x_i \in L$, as correctness of the reduction would imply $K(y) \leq m/2 - g(m)$ with probability $1 - 1/n^{\omega(1)}$ (as we could answer NO for all queries y with $K(y) > m/2 - g(m)$). In the second case, the random variable $f(x_i)$ has entropy at most $m/2 - g(m)/2 + 1$. We show that in this case, with probability $1/n^{\Omega(1)}$, y sampled from $f(x_i)$ has $K(y) < m/2$, which again contradicts the correctness of the reduction. We show that with probability $1/n^{\Omega(1)}$ over y sampled from $f(x_i)$, $K(y \mid x_i) < m/2 - \omega(\log(m))$, from which the previous line follows using Proposition 11 and the fact that $K(x_i) = O(\log(n_i)) = O(\log(m))$. Indeed let k be the entropy of $f(x_i)$ and let $y_1 \dots y_{2^{k+\log(m)}}$ be the first $2^{k+\log(m)}$ strings in the support of $f(x_i)$ in order of decreasing probability, and let Y be the set of these strings. Conditioned on x_i, each such string can be described with a program of size $k + \log(m) + O(\log(m))$, which is $m/2 - \omega(\log(m))$, using the facts that $k \leq m/2 - g(m)/2 + 1$ and $g(m) = \Omega(\log(m))$. We claim that with probability at least $1/m$, y sampled from $f(x_i)$ belongs to Y. Indeed, if not, we have that with probability greater than $1 - 1/m$, y sampled from $f(x_i)$ has probability at most $2^{-(k+\log(m))}$, and hence the contribution to the entropy of $f(x_i)$ from such strings is at least $(1 - 1/m)(k + \log(m)) > k$, which is impossible. Thus we have that with probability $1/m$, y sampled from $f(x_i)$ belongs to Y, and that each string in Y has Kolmogorov complexity smaller than $m/2$. But this contradicts the correctness of the reduction, as x_i is a NO instance, and hence with probability $1 - 1/n^{\omega(1)}$, y sampled from $f(x_i)$ should have $K(y) \geq m/2$ (as otherwise we could answer YES for all queries y with $K(y) < m/2 - g(m)$). □

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